

Project Planning Phase - Script-Controlled ACL

Phase 4 · Project Planning Phase

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This document is the Agile-style project plan for the ServiceNow System Administrator capstone "Script-Controlled ACL - Restrict Record Access Based on Field Value". It captures the product backlog as prioritised user stories and organises the delivery into three sprints, both mapped to the seven implementation milestones for the record-level, field-value-driven access control built on Personal Developer Instance dev443713 using the custom table u_institution_details, roles bb1-bb4, and four Access Control rules.

Project	Script-Controlled ACL - Restrict Record Access Based on Field Value
Track	ServiceNow System Administrator (Naan Mudhalvan / SmartBridge / SmartInternz)
Instance	dev443713.service-now.com (Personal Developer Instance, Australia release)
Team Lead	Arockia Rajamanickam
Team Members	John Richardson Dyriaraj C, Saravana Kumar S, Balaji S, Geethesh B S
Build Date	2026-09-28
Method	Agile - 3 sprints across 7 milestones, verified by impersonation

Milestone Reference (7 Milestones)

ID	Milestone
M1	Create four custom roles: bb1 (read), bb2 (create), bb3 (write), bb4 (delete)
M2	Create test user "EEE User" and assign roles bb1-bb4
M3	Create custom table u_institution_details with the Branch choice field (ECE/EEE/CSE) and student attributes
M4	Populate sample records ECE001, EEE001, EEE002, CSE001
M5	Elevate to security_admin and remove the auto-generated default ACL role u_institution_details_user
M6	Build the four record-level ACLs (READ advanced + condition + script; CREATE bb2; WRITE bb3; DELETE bb4)

ID	Milestone
M7	Verify record-level isolation by impersonation (EEE User, Abel Tuter, System Administrator)

Product Backlog - User Stories

US No	As a / I want to / so that	Acceptance Criteria	Priority
US-1	As a ServiceNow administrator, I want to create four custom roles bb1-bb4 mapped to read/create/write/delete, so that access can be granted one operation at a time.	Roles bb1, bb2, bb3, bb4 exist in sys_user_role, each named for its intended operation.	High
US-2	As a ServiceNow administrator, I want a test user "EEE User" that holds all four bb roles, so that branch-level access can be validated by impersonation.	EEE User exists and holds bb1, bb2, bb3, bb4 (visible in sys_user_has_role).	High
US-3	As a ServiceNow administrator, I want a custom table u_institution_details with a Branch choice field and student attributes, so that institution records can be stored and classified by branch.	Table u_institution_details created with Student Roll Number, Student Name (ref User), Faculty Name (ref User), Branch (choice ECE/EEE/CSE), Email, Phone Number, Description (4000).	High
US-4	As a ServiceNow administrator, I want representative sample records across branches, so that access filtering can be observed against real data.	Records ECE001 (ECE), EEE001 (EEE), EEE002 (EEE), CSE001 (CSE) exist with correct Branch values.	Medium
US-5	As a security administrator, I want to elevate to security_admin and remove the auto-generated default ACL role, so that each ACL depends only on its intended bb role.	security_admin elevation active; role u_institution_details_user removed from the generated ACLs.	High

US No	As a / I want to / so that	Acceptance Criteria	Priority
US-6	As a branch end-user (EEE User), I want to read only records where Branch is EEE, so that I cannot see other branches' data.	READ ACL: role bb1, Advanced, condition u_branch=EEE, script returns true for admin, true for bb1, false otherwise; EEE User sees only EEE001 and EEE002.	High
US-7	As a branch end-user, I want create, write and delete gated by dedicated roles, so that only authorised operations are possible.	CREATE ACL requires bb2, WRITE ACL requires bb3, DELETE ACL requires bb4; no condition and no script on these three.	High
US-8	As a data-security stakeholder, I want the access model verified by impersonation, so that record-level isolation is proven before sign-off.	EEE User sees 2/2 EEE records with New/Update/Delete; Abel Tuter (no bb roles) gets "Security constraints prevent access to requested page"; admin sees all 4.	High

Sprint Plan

Sprint	Milestones Covered	User Stories	Story Points	Duration
Sprint-1: Users, Roles & Table	M1 (roles), M2 (test user), M3 (table + Branch field), M4 (sample records)	US-1, US-2, US-3, US-4	13	2 days
Sprint-2: Access Control Rules	M5 (security_admin elevation, remove default ACL role), M6 (four ACLs + READ script)	US-5, US-6, US-7	13	2 days
Sprint-3: Verification	M7 (impersonation testing: EEE User, Abel Tuter, System Administrator)	US-8	5	1 day
Total Story Points		31		
Total Duration		5 working days		
Definition of Done		All four ACLs active; impersonation shows EEE User limited to 2 EEE records, Abel Tuter blocked, admin sees all 4		